

Appendix

A Experimental Details

A.1 Computing Resources

In this work, we utilized two $8 \times$ NVIDIA A800-SXM4-80GB nodes, each equipped with up to 64 CPU cores and 1 TB of memory; and one $8 \times$ NVIDIA L40-46GB node, equipped with up to 256 CPU cores and 1TB of memory. The nodes were configured to run on Ubuntu 22.04 LTS. This configuration provided the necessary computational power to efficiently process and analyze the data generated during our experiments.

A.2 Experiments Configurations

For all fine-tuning experiments, we employed the AdamW optimizer. The experimental setup is as follows:

- Tables 1 and 2 experiments:
 - During the safety alignment phase, the model was fine-tuned for three epochs with a learning rate of 2×10^{-5} and a batch size of 32. The training process took approximately ten hours on 8 GPUs.
 - In the downstream fine-tuning phase:
 - * For harmful fine-tuning, we trained the model for five epochs using a learning rate of 1×10^{-5} and a batch size of 20. The fine-tuning process took approximately three minutes.
 - * For benign fine-tuning, the model was fine-tuned for three epochs with a learning rate of 2×10^{-5} and a batch size of 64.
- Figure 2 experiments: The model was fine-tuned using a batch size of 20 over five epochs, with a learning rate of 5×10^{-5} .

B High-Similarity and Low-Similarity Subset Selection

Firstly, we obtained representations of both safety alignment and downstream task datasets using a uncensored chat model. Specifically, we employed the Llama 2 (Touvron et al., 2023) base model, which we fine-tuned on the UltraChat dataset (Ding et al., 2023). The rationale for this setup will be discussed in Section 4.1.

Secondly, we computed cosine similarity scores between these representations to quantify their relationships. For each sample in the safety alignment dataset, we calculated the average similarity score by comparing it against all samples in the downstream task dataset. These average similarity scores were used to rank the safety alignment samples.

Lastly, in our experimental framework, we defined two subset sizes (1K and 5K) and selected the top N samples with the highest similarity scores to form the high-similarity subset. Conversely, the bottom N samples with the lowest scores were designated as the low similarity subset. Additionally, a random subset was generated by randomly sampling from all available data points. This methodology enables us to investigate the impact of data similarity on the safety outcomes of fine-tuned models.

C Additional Experimental Results

C.1 Data Contamination Examination

Shi et al. (2024) proposed MIN-K% PROB to examine whether certain data have been seen during training, where an unseen example is likely to contain a few outlier words with low probabilities under the LLM. We then experiment to examine whether such situations are a factor in breaking safety guardrails. As shown in Figure S1, the results indicated that each fine-tuning subset has a low probability of being part of the LLAMA-2-7B-CHAT training data.

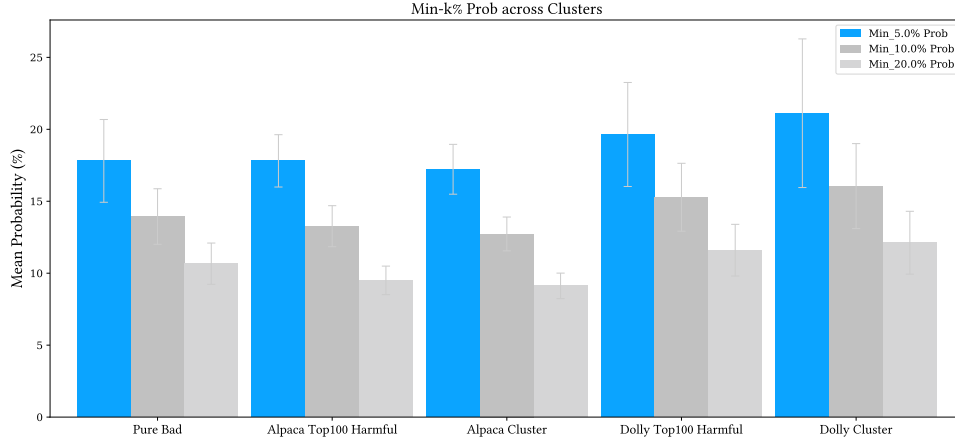


Figure S1: Mean probabilities of membership inference across clusters using the MIN-K% PROB method. The bars represent the average probabilities for different thresholds (5%, 10%, and 20%) across each fine-tuning dataset in Figure 2. Results suggest that each cluster exhibits low inclusion probabilities in the LLAMA-2-7B-CHAT training/alignment data.

C.2 Results on GEMMA-2 2B/9B

We provide our experimental results on GEMMA-2-2B (Table S1) and GEMMA-2-9B (Table S2) (Gem-maTeam, 2024). The results also suggest that the model’s safety guardrail is more durable and resistant when upstream safety alignment data is less similar to the downstream fine-tuning dataset. These results are consistent with our findings on LLAMA-2-7B in Table 1 and LLAMA-2-13B in Table 2.

Safety-alignment Dataset Size (→)		None	Full (7.7K)	5K			1K		
				High-Sim	Random	Low-Sim	High-Sim	Random	Low-Sim
Initial	Utility	7.09	7.11	7.5	7.43	7.21	7.33	6.98	7.32
	HS	70.33%	20.67%	32.33%	24.00%	23.33%	41.67%	40.67%	39.67%
Dataset		Downstream Fine-tuning (Harmful Tasks)							
List	HS	75.33%	71.67%	75.33%	70.00%	69.00%	78.67%	75.33%	65.00%
Pure Bad	HS	85.00%	86.33%	82.67%	82.33%	75.00%	86.67%	86.33%	80.33%
Dataset		Downstream Fine-tuning (Benign Tasks)							
Alpaca	Utility	5.66	5.64	5.14	5.3	5.5	5.52	5.45	5.64
	HS	76.33%	65.67%	76.00%	71.00%	68.00%	80.67%	69.67%	68.33%
SAMSum	Utility	50.35%	51.98%	50.37%	49.81%	50.21%	49.71%	49.60%	50.19%
	HS	75.00%	71.67%	81.67%	79.67%	76.67%	88.33%	84.00%	68.33%

Note. For High-Sim’s and Low-Sim’s Initial models, we report the average score across four target downstream datasets.

Table S1: The Utility/Harmfulness Before/After Downstream Fine-tuning on GEMMA-2-2B.

Safety-alignment Dataset Size ($\rightarrow$)		None	Full (7.7K)	5K			1K		
				High-Sim	Random	Low-Sim	High-Sim	Random	Low-Sim
Initial	Utility	7.95	8.05	8.02	7.83	7.9	7.9	7.84	7.74
	HS	61.33%	8.33%	15.67%	10.67%	10.33%	21.00%	17.67%	14.00%
Dataset	Downstream Fine-tuning (Harmful Tasks)								
List	HS	71.33%	58.00%	64.00%	66.33%	51.67%	60.00%	59.67%	57.33%
Pure Bad	HS	86.33%	89.67%	89.00%	86.33%	85.33%	54.00%	52.67%	48.67%
Dataset	Downstream Fine-tuning (Benign Tasks)								
Alpaca	Utility	5.45	5.35	5.41	5.11	4.84	5.41	5.23	5.51
	HS	78.33%	63.00%	65.67%	66.33%	62.33%	79.00%	81.33%	77.67%
SAMSum	Utility	49.52%	50.82%	49.92%	49.68%	50.23%	49.53%	49.46%	48.50%
	HS	85.00%	52.33%	64.67%	63.67%	58.33%	60.67%	59.00%	58.67%

Table S2: The Utility/Harmfulness Before/After Downstream Fine-tuning on GEMMA-2-9B.